

Note 3.3 Linear Independence

Definitions

- **Definition of Linear Independence**

The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in a vector space V are said to be **linearly independent** if

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n = \mathbf{0}$$

implies that all the scalars $c_1, c_2, \dots, c_n$ must all equal zero.

That is, the linear combination equals to zero vector only have the trivial choice of scalars.

- **Definition of Linear Dependence**

The vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ in a vector space V are said to be **linearly dependent** if there exists scalars $c_1, c_2, \dots, c_n$ not all zero such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n = \mathbf{0}$$

That is, the linear combination equals to zero vector exist nontrivial choices of scalars.

- **Understanding for the Span**

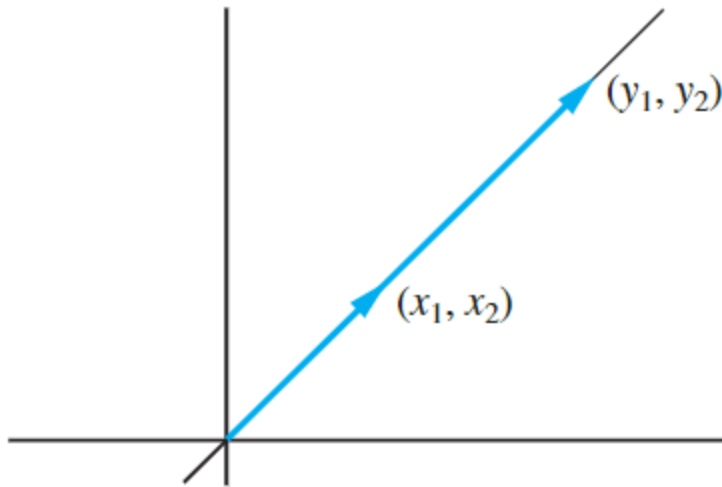
- If $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ span a vector space V and one of these vectors can be written as a linear combination of the other $n - 1$ vectors, then those $n - 1$ vectors span V .
- Given $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, it is possible to write one of the vectors as a linear combination of the other $n - 1$ vectors if and only if there exist scalars $c_1, c_2, \dots, c_n$, not all zero, such that

$$c_1\mathbf{v}_1 + c_2\mathbf{v}_2 + \dots + c_n\mathbf{v}_n = \mathbf{0}$$

that is, they are linearly dependent.

Geometric Interpretation

Suppose $\mathbf{x}, \mathbf{y}$ are two vectors in $\mathbb{R}^2$. If they are linearly dependent, the one can be expressed as the linear combination of the other. That is, they are collinear.



(a) $\mathbf{x}$ and $\mathbf{y}$ linearly dependent

If not, then they are not collinear.

For $\mathbf{x}, \mathbf{y}, \mathbf{z}$ in $\mathbb{R}^3$, if $\mathbf{x}, \mathbf{y}$ are not collinear and they are not zero vector, they span a plane. If the three vectors are linearly dependent, $\mathbf{z}$ lies in the $\mathbf{x}, \mathbf{y}$ -plane. If not, they will span a three-dimensional space.

Theorems and Examples

- Theorem 3.3.1**

Let $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ be n vectors in $\mathbb{R}^n$ and let $X = (\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n)$. The vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ will be linearly dependent if and only if X is singular.

- Proof**

The linear combination of $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n$ equals to zero is

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \dots + c_n\mathbf{x}_n = \mathbf{0}$$

which can be written as the the form

$$X\mathbf{c} = \mathbf{0}$$

where $\mathbf{c} = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{pmatrix}$. Then the equation has a nontrivial solution if and only if X is

singular.

- Application**

To determine whether k vectors $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_k$ in $\mathbb{R}^n$ are linearly independent, we can write

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \dots + c_k\mathbf{x}_k = \mathbf{0}$$

as

$$X\mathbf{c} = \mathbf{0}$$

If $k = n$, that is, X is square, then we can use the determinant to judge whether X is singular.

If $k \neq n$, we need to use the elimination to transform X into the row echelon forms and judge whether there exist free variables.

- **Theorem 3.3.2**

Let $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ be vectors in a vector space V . A vector $\mathbf{v} \in \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ can be written uniquely as a linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ if and only if $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly independent.

- **Proof**

From left to right:

If $\mathbf{v} \in \text{Span}(\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n)$ can be written as the unique linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, that is

$$\mathbf{v} = \alpha_1 \mathbf{v}_1 + \alpha_2 \mathbf{v}_2 + \dots + \alpha_n \mathbf{v}_n$$

suppose that $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$ are linearly dependent, we can write

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n = \mathbf{0}$$

where $c_1, c_2, \dots, c_n$ are not all zero. If we add the two equations together, we will have

$$\mathbf{v} = (\alpha_1 + c_1) \mathbf{v}_1 + \dots + (\alpha_n + c_n) \mathbf{v}_n$$

which is different from the first linear combination and it is conflict with the hypothesis.

So we can prove from left to right.

From right to left:

Suppose there exist more than one linear combination of $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n$, then

$$\mathbf{v} = a_1 \mathbf{v}_1 + a_2 \mathbf{v}_2 + \dots + a_n \mathbf{v}_n$$

$$\mathbf{v} = b_1 \mathbf{v}_1 + b_2 \mathbf{v}_2 + \dots + b_n \mathbf{v}_n$$

Subtract the second equation from the first equation,

$$(a_1 - b_1) \mathbf{v}_1 + (a_2 - b_2) \mathbf{v}_2 + \dots + (a_n - b_n) \mathbf{v}_n = \mathbf{0}$$

here the coefficients are not all zero, which is conflict with the hypothesis. So we can prove from the right to the left.

Vector Spaces of Functions

The Vector Space P_n

We can through the common method to judge whether the polynomials $p_1, p_2, \dots, p_n$ are linearly independent.

The Vector Space $C^{(n-1)}[a, b]$

Determinants can also be used to help to decide whether a set of n vectors is linearly independent in $C^{(n-1)}[a, b]$ (Means the continuous function that has $n - 1$ th derivatives on $[a, b]$).

Let $f_1, f_2, \dots, f_n$ be elements of $C^{(n-1)}[a, b]$, then we have

$$c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x) = 0$$

for each x in $[a, b]$. Then we successively take derivatives of both sides, we have the system

$$\begin{aligned} & c_1 f_1(x) + c_2 f_2(x) + \dots + c_n f_n(x) = 0 \\ & c_1 f_1'(x) + c_2 f_2'(x) + \dots + c_n f_n'(x) = 0 \\ & \vdots \\ & c_1 f_1^{(n-1)}(x) + c_2 f_2^{(n-1)}(x) + \dots + c_n f_n^{(n-1)}(x) = 0 \end{aligned}$$

Then we can use the matrix equation

$$\begin{pmatrix} f_1(x) & f_2(x) & \dots & f_n(x) \\ f_1'(x) & f_2'(x) & \dots & f_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \dots & f_n^{(n-1)}(x) \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

will have the solution $(c_1, c_2, \dots, c_n)^T$. Then we can use this to judge. Here, we can not simply use the [theorem 3.3.1](#) since the function is not identical to the vectors in the matrix.

- Definition of Wronskian**

Let $f_1, f_2, \dots, f_n$ be functions in $C^{(n-1)}[a, b]$, and define the function $W[f_1, f_2, \dots, f_n](x)$ on $[a, b]$ by

$$W[f_1, f_2, \dots, f_n](x) = \begin{vmatrix} f_1(x) & f_2(x) & \dots & f_n(x) \\ f_1'(x) & f_2'(x) & \dots & f_n'(x) \\ \vdots & \vdots & \ddots & \vdots \\ f_1^{(n-1)}(x) & f_2^{(n-1)}(x) & \dots & f_n^{(n-1)}(x) \end{vmatrix}$$

The function is called the Wronskian of $f_1, f_2, \dots, f_n$.

- Theorem 3.3.3**

Let $f_1, f_2, \dots, f_n$ be elements of $C^{(n-1)}[a, b]$. If there exists a point x_0 in $[a, b]$ such that $W[f_1, f_2, \dots, f_n](x_0) \neq 0$, then $f_1, f_2, \dots, f_n$ are linearly independent.

Note that the converse of this theorem is **not valid**.

Example is the function x^2 and $x|x|$ in $C[-1, 1]$ are linearly independent.

And if $f_1, f_2, \dots, f_n$ are linearly independent in $C^{(n-1)}[a, b]$, they will also be linearly independent in $C[a, b]$ since $C^{(n-1)}[a, b] \subset C[a, b]$.

Exercises Section 3.3

Question 5

For **a**,

When $\mathbf{x}_{k+1} \in \text{span}(\mathbf{x}_1, \dots, \mathbf{x}_k)$, that is, $\mathbf{x}_{k+1} = c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + \dots + c_k\mathbf{x}_k$, thus

$$c_1\mathbf{x}_1 + \dots + c_k\mathbf{x}_k - \mathbf{x}_{k+1} = \mathbf{0}$$

That is, $c_{k+1} = -1 \neq 0$. Thus they are linear dependent.

When $\mathbf{x}_{k+1} \notin \text{span}(\mathbf{x}_1, \dots, \mathbf{x}_k)$, then they are linear independent.

For **b**

Suppose the rest vectors are linear dependent, then some vectors can be expressed as the linear combination of the other vectors, which is conflict with the condition that $\mathbf{x}_1, \dots, \mathbf{x}_k$ are linear independent.

Question 12

For **a**,

Since

$$c_1 \cdot 2x + c_2 |x| = 0$$

for all x in $[-1, 1]$, then we can take $x = -1$ and $x = 1$.

$$\begin{cases} -2c_1 + c_2 = 0 \\ 2c_1 + c_2 = 0 \end{cases}$$

Thus $c_1 = c_2 = 0$ and they are linear independent in $C[-1, 1]$.

For **b**,

since $x \in C[0, 1]$, $|x| = x$. Thus

$$c_1 \cdot 2x + c_2 |x| = 2c_1x + c_2x = 0$$

then there exists $c_2 \neq 0, c_1 \neq 0$ satisfying the equation.

Question 18

Let $Y = (\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3)$ and $X = (\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3)$, then since $\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3$ are linearly independent, the matrix Y is nonsingular. Then

$$Y = (A\mathbf{x}_1, A\mathbf{x}_2, A\mathbf{x}_3) = AX$$

since $\det(Y) = \det(A) \det(X) \neq 0$, thus $\det(A) \neq 0$. Then A is nonsingular.

Thus

$$\mathbf{x}_1 = A^{-1}\mathbf{y}_1, \mathbf{x}_2 = A^{-1}\mathbf{y}_2, \mathbf{x}_3 = A^{-1}\mathbf{y}_3.$$

Then for

$$c_1\mathbf{x}_1 + c_2\mathbf{x}_2 + c_3\mathbf{x}_3 = \mathbf{0}$$

we have

$$A^{-1}(c_1\mathbf{y}_1 + c_2\mathbf{y}_2 + c_3\mathbf{y}_3) = \mathbf{0}$$

Then multiply A of both sides, we can know that $c_1 = c_2 = c_3 = 0$ since $\mathbf{y}_1, \mathbf{y}_2, \mathbf{y}_3$ are linearly independent.

(Note the multiplication of matrices. [Multiplication of Matrices](#) It is the linear combination of the column vector)